IBM Hackfest

UniGrid - AI-Orchestrated Middleware for Resilient EV Charging Infrastructure

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Abstract

India’s rapid electric vehicle (EV) adoption has created a **critical gap**: our legacy power grids cannot handle the sudden, **unmanaged loads** of EV charging. When multiple vehicles charge simultaneously, it risks **overloading transformers,** causing **voltage instability (brownouts) and localized blackouts**. Current market solutions are either too expensive, vendor-locked, or lack the intelligence to adapt to India's fluctuating grid conditions.

UniGrid solves this by **retrofitting existing infrastructure** into resilient microgrids. It is a vendor-neutral middleware hosted on **low-cost edge nodes (Raspberry Pi)** that acts as a universal bridge. It integrates modern smart chargers via the **Open Charge Point Protocol (OCPP)** while simultaneously controlling legacy “dumb” chargers through custom smart relays.

Unlike static load balancers, UniGrid uses a dynamic Iterative **Water-Filling Algorithm** to redistribute available power in real-time, ensuring demand never exceeds the grid’s safety limit. Crucially, we integrated a **Hysteresis-based safety loop** that detects voltage dips and preemptively throttles charging to prevent hardware damage. By synchronizing with **local solar generation** and **predicting peak loads** using LSTM models, UniGrid transforms **passive charging points into intelligent, grid-responsive assets**.